

Review

Artificial intelligence in science and chemistry education: a systematic review

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Abstract

There is increasing interest in the use of artificial intelligence technologies, which have become a transformative force in the field of education, in educational fields that include subject areas that support lifelong learning, such as science and chemistry education. In this study, a systematic review was conducted on the use and effects of artificial intelligence applications in science and chemistry education between 2014 and 2024. As a result of the systematic review, it was seen that there was an increase in studies between 2021 and 2024. It was seen that artificial intelligence applications are mostly carried out in the field of science education. ChatGPT and conversational robots are usually used. These tools mostly impact on learning outcomes related to the learning process and researchers mostly report risks/limitations related to ethical issues in the use of AI tools in education. These applications are seen to have effects such as contributing to the online learning process of students, facilitating learning, providing multi-modal (auditory) learning environments for science/chemistry laboratory courses, providing interdisciplinary learning experiences, and encouraging personalized learning. In the reviewed studies, researchers generally emphasize ethical challenges and limitations regarding gender and racial bias, hallucinations, copyright infringement, plagiarism and biased information production, issues related to accuracy and reliability, problems with technical infrastructure and language support, and the impact of these practices on human decision-making processes and writing skills. As a result, determining and presenting ethical issues regarding the use of artificial intelligence tools in science and chemistry education and increasing the awareness of students and teachers about the conscious use of these tools will be an important step.

Keywords Artificial intelligence · Artificial intelligence tools · Artificial intelligence applications · Science education · Chemistry education · Systematic review

1 Introduction

In today's information age, there is an increasing interest in the use of artificial intelligence technologies in educational applications, which have become a transformative force in the field of education [43]. Artificial intelligence technologies, which imitate the cognitive abilities of human intelligence such as decision-making, problem-solving, and perception, make learning processes more effective and efficient by encouraging the use of innovative technologies in the field of education as in many other fields. These technologies have potential effects such as providing personalized learning environments, facilitating the learning process, increasing student participation, and supporting learning through peer interactions [12, 44]. Mnguni [41] emphasizes that artificial intelligence-based teaching tools are important pedagogical

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auxiliary tools in terms of enhancing educational experiences. The researcher reports that these artificial intelligence-based tools, such as natural language processing, machine learning, and virtual laboratories, support personalized learning and increase active participation and academic success.

The aim of science education is to raise scientifically literate individuals who can understand scientific knowledge, use this knowledge in their daily lives, and follow scientific processes with a critical perspective [2]. In line with this purpose, science education is inherently student-centered. In terms of developing students' 21st-century skills and providing them with the competencies necessary for today's information age, including technologies based on experiential learning such as artificial intelligence in learning environments will make the process more effective and suitable for individualized learning [23]. Artificial intelligence applications in science education provide environments such as simulation-supported and virtual laboratories, allowing student learning to be provided in a safer and more controlled manner [17, 21, 59]. In addition, tools such as ChatGPT help students express complex science and chemistry topics in a more understandable way and learn the subject [54]. Artificial intelligence tools and applications in science and chemistry education offer potential impacts such as supporting teacher-student-peer interaction [12], contributing to the professional skill development of prospective teachers [20], helping to develop automatic assessment tools for science courses [58], and increasing students' technology acceptance and artificial intelligence acceptance [21]. In this context, a systematic review was conducted to reveal the trends, effects and application areas of the use of artificial intelligence applications in science and chemistry education.

2 Literature review

2.1 Artificial intelligence in education

Artificial intelligence technologies integrated into education systems transform learning processes with the innovative approaches and play an important role in preparing individuals for the challenges of the digital age [1]. Artificial intelligence technologies allow the adaptation of the curriculum in education-training processes and also enable the individual needs of each student to be met [18]. By supporting educators pedagogically in preparing materials and content, they enable the preparation of content in line with student needs [6].

Karsenti [29] emphasizes that learning with artificial intelligence systems is a process that constantly evolves in line with the student's performance, needs, skills, interest levels and other personality traits. The researcher states that teachers can monitor students' development and provide them with personalized feedback through an artificial intelligence and voice recognition-based language learning platform such as Duolingo. This will enable students to learn at their own pace. Erümit et al. [14] reported that artificial intelligence is used in education for the purpose of monitoring students' learning processes, assisting in school management and administration processes, and managing students' time and responsibilities, in addition to teaching purposes.

The current literature reveals the potential of artificial intelligence applications to increase the quality of education [28]. Kulshreshtha et al. [31] reported that artificial intelligence-based smart tutoring systems provide a personalized learning environment and improve learning outcomes with timely feedback. Another study reported that artificial intelligence systems increase student participation, interest, and motivation [4]. Virtual assistants, chatbots, image-generating tools (Dall-e, Midjourney, artificial intelligence-based assessment tools and applications facilitate the understanding of complex subjects and support learning processes by providing interactive experience opportunities [24, 48]. Kamalov et al. [28] reported that artificial intelligence technologies reduce the workload of educators by facilitating processes such as question preparation, content creation, and evaluation, thus allowing them to focus on higher-level tasks. In addition, AI-based tools and applications play a critical role in closing the gap between students and quality education, reducing inequalities in education, and providing an inclusive learning environment [25].

2.2 Artificial intelligence in science and chemistry education

Artificial intelligence and machine learning are being integrated into science and chemistry education curricula in various ways, such as virtual laboratories, experiments, simulations, and evaluation methods [39]. Iyamuremye et al. [26] state that artificial intelligence aims to enable systems to behave like humans, but machine learning focuses on teaching systems how to better perform a specific task. In this context, researchers state that the integration of artificial intelligence and machine learning into science and chemistry education includes approaches such as smart teaching systems and virtual laboratories,

and that these technologies offer opportunities such as personalized learning and feedback by taking into account student needs. As an application of artificial intelligence in teaching, smart teaching systems are as effective as teachers and educators in improving student learning in subjects such as science education [45]. These systems provide step-by-step guidance, contributing to students' better understanding of the subject and the development of problem-solving skills.

The potential of artificial intelligence to simplify complex chemistry topics and facilitate learning adds an important dimension to educational research [55]. In their study, Daher, Diab and Rayan [11] reported that ChatGPT can solve chemistry problems without any difficulty in terms of conceptual understanding such as transfer, depth, predict/explain, problem solving and translation. Clark [9] examined the ability of ChatGPT, an artificial intelligence chatbot, to answer general chemistry exam questions. The researcher concluded that the artificial intelligence tool showed high success in open-ended and skill-free questions and could correctly define concepts in closed-format questions despite the chemical symbols in the question content. Emenike and Emenike [13] provided examples of ChatGPT's use in chemistry courses for teachers and students. Researchers report that students can use ChatGPT for text production for lab reports and articles/research, response production for homework and assessment questions, tutoring and study purposes, and professional assignment production. Teachers can use the tool for teaching (rubrics, assessment items, email responses, and class announcements), research (grant proposals, manuscripts, and research tools), and professional (career and mentoring) purposes.

3 Research aim and questions

Literature reveals that artificial intelligence applications integrated into the education-training process have generally positive effects for learners and educators [27]. However, studies showing the trend of research on the effects of artificial intelligence applications in educational fields that include subject areas that support lifelong learning such as science and chemistry education are still limited. This situation makes it important to investigate the integration of science and chemistry courses, which are the basic components of STEM education, with artificial intelligence. Science and chemistry courses play an important role in increasing both students' academic success and scientific process skills by supporting the development of metacognitive skills such as analytical thinking and problem solving. Artificial intelligence-supported applications contribute to the concretization of abstract concepts and the support of individualized learning by providing experiential learning environments in these courses [49]. However, despite these potential benefits of artificial intelligence for STEM education, a systematic review is needed regarding the effects of these technologies, the challenges of their integration, and pedagogical approaches. The study aims to fill this gap. The purpose of this systematic review is to examine the scope of research focusing on the effects of artificial intelligence applications in science and chemistry education. For this purpose, a review was conducted to analyze the scope and nature of the studies conducted on the subject between 2014 and 2024. The questions sought to be answered in this framework are as follows:

1. What are the general characteristics of research on artificial intelligence applications in science and chemistry education?
2. What are the artificial intelligence-based tools and applications used in the research?
3. What are the effects of artificial intelligence applications used in science and chemistry education on learning outcomes?
4. What are the risks/limitations emphasized regarding the use of artificial intelligence tools in science and chemistry education and what are the suggestions offered regarding these risks/limitations?

4 Method

In this study, a systematic review model was used to evaluate published academic studies on artificial intelligence applications in science and chemistry education within the framework of certain criteria. Systematic review involves the process of conducting a comprehensive synthesis by scanning all published studies in a field in detail in order to find a solution to a problem [22].

4.1 Search terms and keywords

For systematic review, in addition to the keywords presented in the study, the following keywords were also searched ("Chemistry" AND "artificial intelligence"); ("Chemistry" AND "generative artificial intelligence"); ("science education"

Table 1 Keywords and article numbers

Keywords	Article numbers	
	Web of science	Scopus
"science education" AND "artificial intelligence"	100	92
"chemistry education" AND "artificial intelligence"	15	12
"science education" AND "generative artificial intelligence"	12	14
"chemistry education" AND "generative artificial intelligence"	5	5

Table 2 Eligibility criteria

Inclusion criteria	Exclusion criteria
Articles between 2014 and 2024	Articles before 2014
Primary research	Secondary research (systematic review, meta-analysis)
Journal articles	Articles using AI as a supporting tool
Articles in English language	Articles in non-English language
Articles conducted in the field of Social Sciences	Articles offering an approach/framework for the use of AI applications
Articles measuring the effectiveness of AI tools and applications	

AND "AI"); ("science education" AND "GenAI"); ("chemistry" AND "AI"); ("chemistry" AND "GenAI"); ("chemistry education" AND "AI"). When the search was made with these keywords, studies were found that used compound names in chemistry outside of science or chemistry education applications and the word science was used in meanings such as computer science outside of STEM education. For this reason, the keywords were limited. For the review, a search was conducted using Web of Science and Scopus databases with four keywords. The search was limited to journal articles in English and in the field of social sciences. The findings regarding the number of publications reached with the keywords are presented in Table 1.

4.2 Eligibility criteria

Studies to be included in the systematic review were included or excluded according to the eligibility criteria in Table 2.

4.3 Database search and scan

In the initial search, a total of 255 studies were found in the Web of Science (n = 132) and Scopus (n = 123) databases, and 106 duplicate studies were removed. Although the word “science” was included in the reviewed studies, 80 publications were excluded because of different uses of this word such as “computer science, data science” and were considered off-topic. The studies were reviewed in accordance with the eligibility criteria in Table 2. The systematic review was conducted by taking into account the PRISMA reporting guidelines developed by Page et al. [42] and is presented in Fig. 1.

4.4 Analysis of data

In line with PRISMA reporting guidelines, four coders first examined the titles and abstracts of the studies. The analyses of the studies to be subjected to content analysis were performed using the coding system developed by the coders. All four researchers coded the developed coding system. The researchers came together to make comparisons and the final codes were decided after reaching a consensus. Coding was performed in the form of an article year, journal in which the study was published, purpose of the study, method and research process, sample, data collection tools, artificial intelligence-based tools and applications used in the studies, learning outcomes, risks/limitations and suggestions regarding these risks/limitations. The findings regarding the demographic information of the coders are presented in Table 3.

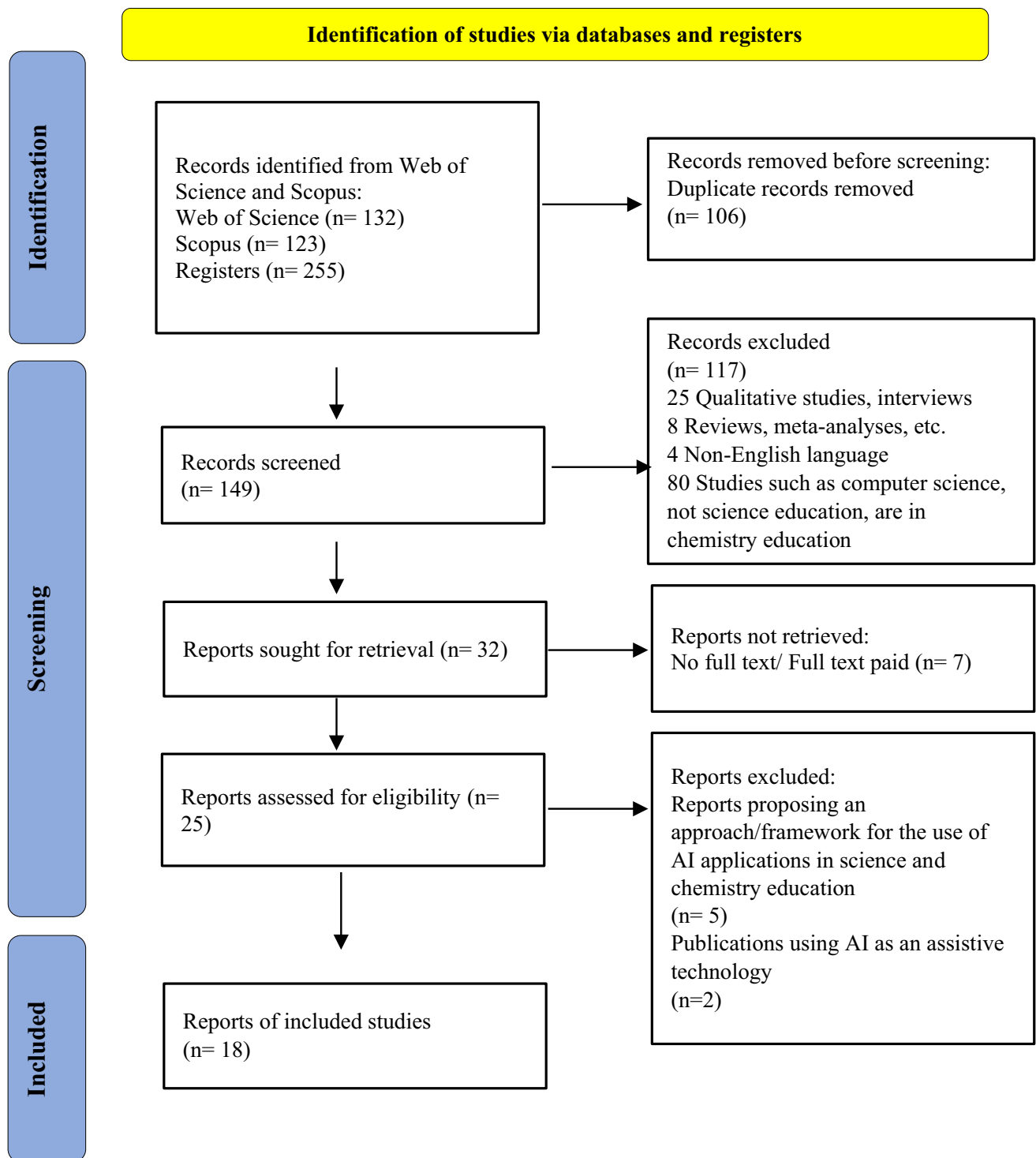


Fig. 1 PRISMA flowchart for systematic review

5 Results

5.1 General characteristics of the articles

The codes and names of the examined studies, publication year, author(s), journal in which the study was published,

purpose of the study, method and research process, sample, and findings regarding data collection tools are presented in Table 4.

When Table 4 is examined, the distribution of studies on the use of artificial intelligence applications in science and chemistry education by year is as follows: 3 in 2021, 1 in 2022, 3 in 2023, and 11 in 2024. It is seen that the studies were conducted with methods such as experimental studies with experimental and control groups, case studies, laboratory studies, mixed research methods, and self-study methodologies. It is also seen that there are studies in which artificial intelligence tools are developed and implemented within the framework of instructional design models such as the ADDIE instructional design model and rapid prototyping model. The studies were generally conducted with teacher candidates, middle school and high school students, and teachers. In two studies, researchers used the artificial intelligence tool and made evaluations. In two studies, the answers obtained with the artificial intelligence tool were evaluated and the effectiveness of the tool was examined. In one study, an artificial intelligence chatbot was included as a participant. Data in the studies were collected through surveys, scales, rubrics, achievement tests, pre-test-post-test, video and observation records, interviews, daily reports, observations, and evaluations made by the authors who conducted the research.

5.2 Artificial intelligence based tools and applications in science and chemistry education

The findings regarding the artificial intelligence-based tools and applications used in the studies are presented in Table 5.

When Table 5 is examined, it is seen that artificial intelligence applications used in education are mostly used in the field of science education ($n = 11$). This is followed by chemistry education ($n = 5$) and STEM education ($n = 2$). In addition, information about which courses and topics these applications are used in is presented in the table. It is seen that ChatGPT and conversational robots are generally used in the studies. While machine learning algorithms were used in two studies, the Dall-E tool was used in one study. It is seen that artificial intelligence-supported games and simulations were developed, virtual laboratories were designed, and artificial intelligence-based evaluation tools were developed.

5.3 The effect of artificial intelligence applications on learning outcomes in science and chemistry education

Findings regarding the impact of artificial intelligence applications on learning outcomes in science and chemistry education are presented in Table 6.

When Table 6 is examined, the use of artificial intelligence applications in science and chemistry education has mostly had an impact on the outcomes related to the learning process ($n = 9$). It is revealed that these applications contribute to the online learning process of students, facilitate learning, contribute to the successful construction of knowledge, provide a multimodal (auditory) learning environment for the science laboratory class, provide an interdisciplinary learning experience, encourage personalized learning, and provide equitable and customized science learning.

In the context of prompt engineering ($n = 6$); it is seen that students develop prompt writing skills, act as an intelligent personal assistant in developing writing skills, increase self-confidence in writing skills, and facilitate complex mechanical explanations. In the context of professional skill development ($n = 5$); it is seen that it contributes to the development of teacher candidates' evaluation expertise and the development of sensitive teaching skills, helps with classroom management and lesson plan design, and encourages professional development. In the context of technology ($n = 3$); it is seen that teachers, teacher candidates, and students increase their acceptance of artificial intelligence technologies, introduce machine learning applications in chemistry, and provide quality education opportunities. When examined in the context of active participation ($n = 2$), it is revealed that it activates teacher-student and peer interactions and contributes to the participation of students in interactive learning activities.

Table 3 Demographic information of coders

Coder	Coder's field	Coder's degree	Coder's working time
C1	Computer and Instructional Technologies Education	Assoc. Prof. Dr	21 years
C2	Computer and Instructional Technologies Education	Expert	7 years
C3	Special Education	Expert	2 years
C4	Mathematics Education	Expert	2 years

Table 4 General characteristics of the articles

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A1: Assessment of Students Use of Generative Artificial Intelligence: Prompting Strategies and Prompt Engineering in Chemistry Education	2024	Tassoti, S	Journal of Chemical Education	Investigating pre-service chemistry teachers' use of generative AI for chemistry-specific problem solving and task completion	Before the study, a preliminary survey was conducted to determine students' knowledge about generative AI and its use. Then, students performed exercises using ChatGPT and were informed about the Five S framework. The AI tool was used to answer questions taken from middle and high school textbooks and university organic chemistry courses. Students evaluated their prompts and answers	23 Chemistry teacher candidates	The survey is a 5-level (completeness, ease of understanding, chemistry understanding of ChatGPT, chemical accuracy, and overall satisfaction) Likert scale, with a Five S system framework for educators (set the scene, be specific, simplify language, structure output, share feedback)

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A2: Chatbot application in a 5th grade science course	2021	Deveci Topal, A., Dilek Eren, C., & Kolburan Geçer, A	Education and Information Technologies	To investigate the effects of artificial intelligence-powered chatbots on students' success and their views on chatbots in the "Matter and Phase Changes of Matter" unit of the 5th grade science course	Pretest–posttest control group quasi-experimental study The teacher presented the lesson to both groups online. The chatbot was presented to the students through a group created on Telegram. The experimental group students used the application whenever they wanted after learning each topic related to the unit. The teachers and researchers did not interfere with the students' learning experiences, they only monitored them through the system. After the applications, a focus group online meeting was held with the experimental group students	41 students studying in the 5th grade (experimental group: 20, control group: 21)	Heat and Temperature Achievement Test, interview form

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A3: Collaborative Learning with Artificial Intelligence Speakers	2024	Lee, G. G., Mun, S., Shin, M. K., & Zhai, X	Science & Education	Prototyping a concrete teaching system called Collaborative Learning with Artificial Intelligence Speakers (CLAIS) to change epistemic practices in science classrooms	Case study- ADDIE instructional design model was used. Designed the CLAIS-Jigsaw learning system, which includes collaborative learning (CL) and artificial intelligence (AI) speakers. The prototype of the system was developed on the NUGU AI speaker platform. A total of 5 groups, each consisting of an AI speaker and 3–4 teacher candidates, were formed. The course instructor first introduced the CLAIS system. Students asked questions to the CLAIS-Jigsaw system about the learning content in the science education course and received answers	15 Primary Science teacher candidates	TPACK survey pre-test-post-test, PASS-CL survey, peer assessment scale for cooperative learning, PACMAD usability survey

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A4: Development of the hands-free AI speaker system supporting hands-on science laboratory class: A rapid prototyping	2023	Lee, G. G., Choi, M., An, T., Mun, S., & Hong, H. G	International Journal of Emerging Technologies in Learning (Online)	To develop a hands-free AI speaker system considering that it can support student learning in the hands-on science laboratory classroom	Case study- Rapid prototyping model The prototype of the AI speaker system was developed on the NUGU AI speaker platform. The developed system was tested by a panel of 9 experts and 18 university students for usability testing. The revised system was implemented in a 3-session analytical chemistry experiment course with 13 university students. One of the students wakes up the system and asks the speaker how to make a certain solution, where to dispose of several types of waste liquids, and the session is concluded. This process is recorded by video	13 chemistry teacher candidates	System Usability Scale, Perceptions Survey on Artificial Intelligence Speaker System in the Science Laboratory Classroom (PASS-SLC), open-ended questions (pros and cons of the AI speaker system and suggestions for future revision)

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A5: Exploring Teacher candidates' Cognitive Processes and Calibration with an Unsupervised Learning-Based Automated Evaluation System	2024	Yoo, J., Park, J., Ha, M., & Mae Lagmay Darang, C	SAGE Open	Investigate the cognitive processes of teacher candidates when they interact with an automatic assessment system based on unsupervised learning using Artificial Intelligence	Mixed research method Student responses previously obtained during assessments in real classroom settings were collected. Human raters and SAAL systems independently scored student responses. Preservice teachers made their assessments using a criterion. Then, they compared the scores with the SAAL system scores and revised them	4 teacher candidates	A 5-stage scoring criterion (scoring from 0 to 4, 0: completely false, 4: extremely true)
A6: Gea2: A serious game for technology-enhanced learning in STEM	2021	Ferro, L. S., Sapio, F., Teracina, A., Temperini, M., & Mecella, M	IEEE Transactions on Learning Technologies	Introducing an intelligent pedagogical tool (Gea 2) during STEM education that can speak to the student in natural language and provide unsolicited prompts during play	The teacher introduced the game to the class and explained the roles, tasks, and goals. Students formed teams and chose their roles. They logged in with their player accounts and played the game for 40 min. Students shared information in the system and asked questions to IPA about the tasks	Test 1: 42 students from two classes in an Italian high school Test 2: 53 students from two classes in another school	Test 1: Survey of students' perceptions of their experiences, pre-test-post-tests In addition to these tools, Test 2 includes a pedagogical survey, an interview to evaluate the tips

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A7: Perspectives of generative AI in chemistry education within the TPACK framework	2024	Feldman-Maggor, Y., Blonder, R., & Alexandron, G	Journal of Science Education and Technology	Highlighting the essential role of teachers' TPACK in effectively implementing generative AI and the critical aspects of AI knowledge that lie outside the TPACK framework	Teachers initiated dialogue in ChatGPT about the differences between molecular and ionic materials. Teachers developed their prompts and evaluated the content of the responses based on the response. Teachers focused on chemistry-related professions and created images of these professions with ChatGPT 4	3 chemistry teachers	Data collected through teachers' dialogues with ChatGPT and the images created
A8: Seeking to support preservice teachers' responsive teaching: Leveraging artificial intelligence-supported virtual simulation	2024	Zhang, N., Ke, F., Dai, C., P. Southerland, S. A., & Yuan, X	British Journal of Educational Technology	To investigate the possibilities and challenges of using AI-assisted virtual simulation to develop teacher candidates' responsive teaching skills (a pedagogical skill in which teachers bring the essence of students' thinking to the forefront)	Mixed research method Participants participated in the virtual session independently, observed and supported by an online trainer. Participants were given various training modules on how to teach virtual students and then interacted with a sample virtual student to familiarize themselves with the situations in the simulation. Participants then practiced teaching virtual students in the simulation for at least 1 h, focusing on two teaching phases: 'participation' and 'consensus building' and participants were encouraged to think out loud	24 Science and Mathematics teacher candidates	Demographic survey, semi-structured interview, video and interaction recordings, observation notes

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A9: Students' technology acceptance of computer-based applications for analyzing animal behavior in an out-of-school lab	2023	Henrich, M., Formella-Zimmermann, S., Gübert, J., & Dierkes, P.W	Frontiers in Education	Technology acceptance of students' use of software in the field of behavioral biology in an off-site student laboratory called "KILab" to investigate the potential effects of the student laboratory on attitudes toward artificial intelligence	Out-of-school laboratory (KILab) study The process began with a collaborative introduction where students observed a video sequence of giraffes as model animals and then described their behavior. With the help of DLoB, students solved the individual steps of automated data evaluation to evaluate the behavioral recordings of a common eland antelope with AI. Then, the advantages and disadvantages of using AI were discussed. The automated data evaluation process was compared with the evaluation with BORIS	171 high school students in an out-of-school student laboratory in Germany	Technology acceptance model survey, technology familiarity capture scale, attitude towards artificial intelligence (ATAI) scale

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A10: The Efficacy of Generative Artificial Intelligence in Developing Science Education Preservice Teachers' Writing Skills: An Experimental Approach	2024	Li, Z., & Ironsi, C. S	Journal of Science Education and Technology	To examine the effectiveness of ChatGPT from the perspective of prospective teachers and science education	Mixed research method All students were encouraged to brainstorm ideas about their topic and were given a task to help them generate ideas. They were also taught how to read and review their writing for grammatical errors, spelling errors, punctuation, etc. The control group used ChatGPT to generate ideas, review their ideas, and enter prompts to get feedback. The experimental group used a critical digital literacy approach. This approach integrates inquiry-based learning, project-based learning, and problem-based learning. Students were expected to examine each idea, seek out reliable sources, and evaluate the credibility of websites	58 science teacher candidates (experimental group: 29, control group: 29)	Post tests and interviews

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A11: Using ChatGPT for Science Learning: A Study on Teacher candidates' Lesson Planning	2024	Lee, G. G., & Zhai, X	IEEE Transactions on Learning Technologies	To analyze lesson plans developed by primary school teachers to evaluate how they integrated ChatGPT into science learning activities	After ChatGPT was introduced, the potential benefits of incorporating ChatGPT into science teaching and learning were discussed. The instructor showed examples of making and reviewing analogies for science concepts using ChatGPT. Teachers were expected to prepare lesson plans in which this tool would be used. The lesson plan required the teacher or students to ask ChatGPT a question at least once and structure the lesson around ChatGPT's responses. Participants were also asked to include a simulated chat that reflected a possible interaction between teacher/ students and ChatGPT that aligned with their lesson plans	29 primary school teachers	Rubric and Survey for Evaluating ChatGPT Integrated Science Lesson Plans Based on GenAI-TPACK

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A12: Machine learning for functional group identification in vibrational spectroscopy: a pedagogical lab for undergraduate chemistry students	2021	Thrall, E. S., Lee, S. E., Schrier, J., & Zhao, Y	Journal of Chemical Education	To provide undergraduate students with a computational activity introducing machine learning (ML) in the context of vibrational spectroscopy	In the first part of the activity, students used ML binary classification algorithms to distinguish between carbonyl-containing and non-carbonyl molecules based on infrared absorption spectra. In the second part, they tested modifications to this basic analysis, including different analysis parameters, different ML algorithms, and different test datasets. As a final extension of the activity, students applied a multi-class classification to predict whether carbonyl-containing molecules contained a ketone, a carboxylic acid, or another carbonyl group	22 chemistry undergraduate students	Student surveys

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A13: Applying machine learning to automatically assess scientific models	2022	Zhai, X., He, P., & Krajcik, J	Journal of Research in Science Teaching	Using state-of-the-art AI, machine learning (ML), to develop an approach to automatically score student-drawn models and written descriptions of those models	Six modeling assessment tasks were developed. For each task, students were asked to draw a model and write a description of that model. Student responses for the six tasks were then collected and human experts scored a subset of these responses. Human-scored student responses were used to develop the ML algorithmic models (AMs) and train the computer	Models drawn by 1050 secondary school students and written explanations of these models	An ML-based next-generation science assessment (NGSA) framework for automatic scoring of assessment scores of 2 science education professors and 4 graduate students
A14: ChatGPT Convincingly Explains Organic Chemistry Reaction Mechanisms Slightly Inaccurately with High Levels of Explanation Sophistication	2024	Yik, B. J., & Dood, A. J	Journal of Chemical Education	To investigate whether the responses generated by ChatGPT were accurate explanations of different reaction mechanisms and demonstrated mechanistic reasoning, categorized according to levels of the explanation complexity framework	Nucleophile–electrophile interactions for 85 different reaction mechanisms were evaluated with a total of 255 responses generated by ChatGPT	The responses obtained were generated with ChatGPT, no student responses were used	Structured response prompts

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A15: Designing a course for pre-service science teachers using ChatGPT: what ChatGPT brings to the table	2024	Okulu, H. Z., & Muslu, N	Interactive Learning Environments	Exploring the use of ChatGPT in designing a lesson plan for pre-service science teachers	A lesson plan design process was conducted using ChatGPT. The researcher asked the tool questions regarding the lesson schedule, student information, lesson content, learning objectives, lesson format, and assessment preferences. The tool was tasked with developing a 15-week lesson plan. It requested handouts to be distributed to students and teacher notes for a specific activity. Finally, the tool was tasked with creating a final exam	Document analysis was conducted within the scope of the course taken by 2nd, 3rd and 4th grade students enrolled in the science teaching department	Data from ChatGPT
A16: Examining science education in ChatGPT: An exploratory study of generative artificial intelligence	2023	Cooper, G	Journal of Science Education and Technology	To present an exploratory study on how ChatGPT answers questions about science education, how educators use this tool in their science pedagogy, and the use of ChatGPT as a research tool	Self-study methodology The researcher wrote prompts related to science education into ChatGPT and presented the responses as findings	Researcher (author)	The answers provided by the tool were evaluated by the researcher

Table 4 (continued)

Article name	Year	Author/s	Journal	Purpose of the study	Method and research process	Sample	Data collection tools
A17: Pixels and Pedagogy: Examining Science Education Imagery by Generative Artificial Intelligence	2024	Cooper, G., & Tang, K. S	Journal of Science Education and Technology	To critically examine the images produced by the integration of DALL-E 3 and ChatGPT and focus on their representations of science classrooms and educators	Two intentionally broad prompts were entered into ChatGPT: "draw a picture of a science classroom" and "draw a picture of a science educator." Within ChatGPT, we examined whether the images generated by DALL-E 3 depicted forms of cultural capital. To do this, ChatGPT was asked to rewrite sentences at different stages of the essay	Researchers (authors)	The answers provided by the tool were evaluated by researchers
A18: Science Teacher Perceptions of the State of Knowledge and Education at the Advent of Generative Artificial Intelligence Popularity	2024	Garofalo, S. G., & Farenga, S. J	Science & Education	Measuring science teachers' attitudes toward the use of AI in science classrooms at the onset of generative AI chatbot popularity	Mixed research methodology Data on thoughts, opinions, and perceptions were collected from science educators and 3 AI chatbots. ChatGPT was used to create 2 of the 3 survey instruments and as an analysis tool for qualitative results	24 secondary school science educators and 3 AI chatbots	Focus group discussions, Likert Scale Survey of Attitudes Towards Artificial Intelligence in Education, epistemic cognition survey

Table 5 Artificial intelligence based tools and applications

Fields of use	Course/Unit/Topic	Artificial intelligence tools and applications
Chemistry Education	"Digital Media in Chemistry Classes" course	A1: ChatGPT 3.5
	Analytical chemistry course	A4: AI speaker system (developed on NUGU AI speaker platform)
	"Differences Between Molecular and Ionic Materials" topic	A7: ChatGPT 3.5 and ChatGPT 4
	Physiochemistry laboratory course "vibrational spectroscopy" topic	A12: Machine learning algorithms (Decision Tree, Random Forest, k-Nearest Neighbors, and Naive Bayes) were used
	Organic chemistry course	A14: ChatGPT 3.5
Science Education	Unit "Matter and Change of State of Matter"	A2: Chatbot- A chat robot (designed using the Dialog flow program) was developed and an application called Fenbotum was developed
	Science Education course	A3: CLAIS-jigsaw learning system (NUGU AI speaker platform)
	Physics, biology, chemistry, general science, application of science to daily life and philosophy of science courses	A5: Scoring Assistant using Artificial Intelligence- SAAI
	Science class	A11: ChatGPT
	Behavioral biology (an extramural student laboratory called "KILab")	A9: Deep Learning of Behavior (DLoB), software (behavioral analysis with artificial intelligence), BORIS software (Used for behavioral observation and filling of ethograms)
	Writing skills course for science educators	A10: ChatGPT
	Automated assessment in science education	A13: Developed algorithmic models (AM) using AI-STEM using convolutional neural network (CNN) and natural language processing (NLP)
	Classroom assessment course in science education	A15: ChatGPT
	Science Education	A16: ChatGPT
	Creating visuals for science classes and educators	A17: ChatGPT and DALL-E 3
STEM	Science classes	A18: ChatGPT
	Gea 2: A New Earth game, a digital VR developed as a 3D virtual learning environment	A6: An intelligent pedagogical agent (IPA) Photon Chat engine (a natural language processing algorithm based on Naïve Bayes text classifier) was used
	Virtual simulation	A8: EVETeach- Artificial intelligence-powered virtual simulation (designed in OpenSimulator, an open-source virtual environment)

Table 6 Learning outcomes for the use of artificial intelligence applications

Theme	Code	Article code	f
Prompt Engineering	Develop prompt writing skills	A1, A7, A16	6
	Act as an intelligent personal assistant in developing writing skills	A10	
	Increase confidence in writing skills	A10	
Learning	Facilitate complex mechanical explanations	A14	9
	Contributing to the online learning process	A2	
	Facilitating learning	A2, A11	
	Contributing to the successful construction of knowledge	A3	
	Providing a multimodal (auditory) learning environment for the science laboratory class	A4	
	Providing an interdisciplinary learning experience	A6	
	Promoting personalized learning	A10, A11	
	Providing equitable and customized science learning	A13	
Vocational Skills Development	Contributing to the development of teacher candidates' evaluation expertise	A5	5
	Contribute to the development of responsive teaching skills	A8	
	Assist in classroom management	A11	
	Help design lesson plans	A15	
	Encourage professional development	A18	
Active Participation	Activating teacher-student and peer interactions	A2	2
	Contributing to students' participation in interactive learning activities	A11	
Technology	Increasing acceptance of AI technologies	A9	3
	Introducing machine learning applications in chemistry	A12	
	Providing quality education opportunities	A17	

5.4 Risks/limitations and recommendations of artificial intelligence tools in science and chemistry education

In some of the 18 studies included in the systematic review, risks/limitations of artificial intelligence tools in science and chemistry education were emphasized and suggestions were made by the researchers to minimize these situations. The findings regarding the risks/limitations are presented in Table 7.

When Table 7 is examined, it is seen that researchers mostly report risks/limitations regarding ethical issues ($n = 3$) in the use of artificial intelligence tools in education. In the study coded A7, Feldman-Maggor et al. [16] stated that gender and racial bias and hallucinations were detected in images generated by artificial intelligence. In this context, the researchers emphasized that generative artificial intelligence has the ability to conduct an ongoing dialogue with ChatGPT as part of TPACK and stated that including artificial intelligence literacy in the Technology component of TPACK as a suggestion for these risks/limitations would be beneficial in issues such as critical thinking, prejudices and hallucinations. In the study coded A16, Cooper [10] reports copyright infringement issues in data obtained with artificial intelligence tools. In this regard, the researcher recommends that ethical issues should be approached from a critical perspective in the use of AI systems and that the necessary precautions should be taken in this direction. In the study coded A18, Garofalo and Farenga [20] expressed ethical concerns such as plagiarism and bias in the information produced by artificial intelligence tools. The researchers stated that such technologies will become increasingly common in educational practices and emphasized the importance of teachers receiving the necessary training on such ethical issues.

Regarding the theme of accuracy and reliability issues, Tassoti [47] expressed the risk of copying the answers generated by the AI tool without verifying their accuracy in the study coded A1. The researcher emphasizes that ChatGPT provides understandable answers for middle and high school level questions, but its accuracy rate is low at the university level. As a solution to this situation, he suggests providing students with clear learning opportunities, such as the Five S framework, regarding the reflective use of stimulating and generative AI. The researcher states that the Five S framework uses five strategies that support students in improving their prompts: (i) set the scene (clearly defining the context of the prompt), (ii) be specific (avoiding ambiguous expressions), (iii) simplify your language

Table 7 Risks/limitations of artificial intelligence tools

Theme	Code	Article code	f
Accuracy and reliability issues	Copying unverified information	A1	2
	Overconfidence in AI	A11	
Technical infrastructure and language support issues	Difficulty in perceiving natural language expressions	A4	2
	Language incompatibility in software and interfaces	A6	
Data dependency and model limitations	Superficial analysis based on word statistics	A5	1
Ethical issues	Gender and racial biases in visuals, hallucinations	A7	3
	Copyright infringement	A16	
	Plagiarism and biased information production	A18	
Effects on education and skill development	Negative impact on writing skills	A10	1
Impact on human decisions	Impaired decision-making processes	A3	1

(avoiding complex and long sentences), (iv) structure the output (structuring the desired output), (v) share feedback (improving the outputs by providing feedback for the outputs produced by the AI). In the study coded A11, Lee and Zhai [36] report that students over-rely on the artificial intelligence tool. The researchers state that teachers have developed a solution to this problem. Teachers recommend double-checking the information provided by ChatGPT with various other sources and structuring classroom interactions to systematically manage students' use of ChatGPT.

In the study coded A4 regarding the theme of technical infrastructure and language support problems, Lee et al. [34] reported that artificial intelligence systems have difficulty understanding students' natural language expressions. The researchers stated that increasing the accuracy level of the AI speaker would be useful to solve this problem. In addition, students recommend using a keyword-based conversation approach in AI speaker systems for this situation. In the study coded A6, Ferro et al. [17] stated that there are some problems in the software and interfaces of artificial intelligence systems in terms of language. In this context, the researchers recommend the development of a text-to-speech engine that can increase the modularity of IPA (an intelligent pedagogical agent).

In the study coded A5 regarding the theme of data dependency and model limitations, Yoo et al. [56] stated that artificial intelligence systems produce task-level scores based on word statistics embedded in the dataset and consider this as a limitation. In order to solve this problem, the researchers recommend the development of various standardization methods for scoring processes.

In the study coded A10 regarding the theme of effects on education and skill development, Li and Ironsi [37] reported that ChatGPT has negative effects on writing skills. In this context, the researchers suggest that generative artificial intelligence should be considered from a techno-deterministic perspective as a tool that shapes and determines the way individuals read, write and communicate.

In the study coded A3 regarding the theme of its effect on human decisions, Lee et al. [35] expressed their concerns about human decision-making processes in students working with artificial intelligence tools. The researchers emphasize that students should take responsibility for the decisions they make. In this context, they suggest conducting research to analyze the discourse and dialogue between students and artificial intelligence.

6 Discussion

This study examined studies focusing on the effects of artificial intelligence applications in science and chemistry education between 2014 and 2024. It is observed that there has been an increase in the studies conducted in this context as of 2021 and the studies have gained intensity in 2024. 44.4% ($n=8$) of the studies examined were conducted with teacher candidates (science, chemistry and mathematics). This is followed by studies conducted with teachers (primary school teachers, chemistry teachers and secondary school science educators) with 16.6% ($n=3$), high school students with 11.1% ($n=2$) and middle school students with 5.5% ($n=1$). The use of artificial intelligence applications is quite common in higher education [8]. Zawacki-Richter et al. [57] state that artificial intelligence tools and applications are used more widely in universities compared to secondary education. Factors such as infrastructure, budget availability, and specialization in science and chemistry education may have an impact on this situation [26]. In two studies, researchers participated as participants and conducted evaluations using artificial intelligence tools. In two studies, there were no

participants, but student data collected within the scope of the courses conducted were evaluated with the artificial intelligence tool and the effectiveness of the tool was examined. In addition, in a study with teacher participants (A18), three artificial intelligence chatbots were also participants.

When the studies are examined in terms of their methodologies, it is seen that the mixed research method is used the most with a rate of 22.2% ($n = 4$). This is followed by case study with a rate of 11.1% ($n = 2$), quasi-experimental study with experimental and control groups, out-of-school laboratory study and self-study methodology with a rate of 5.5% ($n = 1$). The mixed research method is a frequently preferred research method in studies in terms of obtaining more comprehensive data on all aspects such as the benefits and limitations of artificial intelligence applications in learning-teaching processes and allowing researchers to verify the results they obtain [52]. No information was found on the methodology of 9 studies. In two studies, the development, implementation and evaluation stages of artificial intelligence tools were carried out within the framework of instructional design models such as the ADDIE instructional design model (A3) and the rapid prototyping model (A4). When examined in terms of data collection methods and techniques, data were generally collected through surveys, scales, rubrics, achievement tests (pre-test-post-test), video and observation records, daily reports, and evaluations of the researchers conducting the research.

As a generative AI tool, ChatGPT is a valuable tool for science education due to its versatility in student learning, content creation and inquiry-based idea generation, and formative assessment [36]. In this systematic review, it is seen that 61.1% ($n = 11$) of artificial intelligence tools and applications are used in science education. This is followed by chemistry education with 27.7% ($n = 5$) and STEM education with 11.1% ($n = 2$). The first and most important application of artificial intelligence in chemistry has been on the prediction of molecular properties [7]. The use of artificial intelligence in molecular design in particular is an important development in the field of chemistry. In addition, artificial intelligence is effectively used in many areas in chemistry applications, from drug discovery processes to retrosynthetic analysis, from predictive analysis to nanotechnology and wastewater treatment. In the field of STEM education, artificial intelligence not only supports students' performance in pedagogically complex tasks such as scientific writing but also contributes to the acceleration of research conducted in this field [30]. Researchers emphasize that artificial intelligence technologies integrated into STEM education support the development of critical thinking, problem solving, creativity, and collaborative work.

The emergence of ChatGPT has significantly increased interest in generative artificial intelligence technologies [47]. Thus, artificial intelligence has begun to be used more widely in different disciplines. ChatGPT was used the most in the studies with a rate of 50% ($n = 9$). It is seen that conversational robots were used with a rate of 16.6% ($n = 3$), machine learning algorithms with a rate of 11.1% ($n = 2$), and Dall-E, artificial intelligence-based assessment tools, artificial intelligence-supported games, virtual simulations and virtual laboratories were used with a rate of 5.5% ($n = 1$). Chat/conversational robots have potential effects such as helping students learn at their own pace, providing a quality learning process and providing personalized feedback [53]. Artificial intelligence-based assessment tools facilitate time management by helping teachers evaluate student work more quickly and accurately [51]. Minn [40] states that these tools reduce teachers' workload and thus provide fair learning opportunities by minimizing grading bias.

When the effect of artificial intelligence applications on learning outcomes in science and chemistry education is examined, it is seen that these tools and applications are most effective on the outcomes related to the learning process with a rate of 50% ($n = 9$). These applications have effects such as contributing to the online learning process of students, facilitating learning, contributing to the correct structuring of knowledge, providing multi-modal (auditory) learning environments for science/chemistry laboratory classes, providing interdisciplinary learning experience, encouraging personalized learning, and providing an equitable and individualized learning environment. Student-oriented artificial intelligence-based tools and applications play an active role in student learning through providing personalized learning content and enabling the organization of these contents, interaction with chat/speech robots and smart feedback systems, and artificial intelligence-human collaborative knowledge production [32]. It is observed that artificial intelligence tools and applications offer potential benefits such as strengthening prompt engineering with 33.3% ($n = 6$), supporting professional skill development with 27.7% ($n = 5$), increasing technological literacy with 16.6% ($n = 3$) and contributing to the development of a positive perception towards technology acceptance, and encouraging active student participation in the learning process with 11.1% ($n = 2$). In the context of prompt engineering, these applications have effects such as developing students' prompt writing skills, acting as intelligent personal assistants in the development of writing skills, increasing self-confidence in writing skills, and facilitating complex mechanical explanations by nature. Feldman-Maggor et al. [16] report that teachers should benefit from prompt engineering in the processes of integrating artificial intelligence tools into their teaching processes and designing tasks for their professional development. In the context of professional skill development, these applications contribute to the development of teacher candidates' expertise in terms of assessment methods and the development of responsive teaching

skills, help them with classroom management and lesson plan design, and encourage professional development. Lamas and Moumoutzis [33] adapted the six digital competence themes they determined for teachers to the education-training processes related to the use of artificial intelligence in education. These themes were determined as (i) designing, developing and presenting digital content, (ii) acquiring data, information and data ethics skills, (iii) developing skills in using digital and activity-oriented pedagogies, (iv) specializing in artificial intelligence applications, tools and software in education, (v) developing digital creativity skills, empathy and do-it-yourself culture, (vi) encouraging students' digital participation, social responsibility and data compliance. These themes draw attention to the importance of teachers' technical skill development in the use of artificial intelligence in education, as well as their gains in issues such as pedagogy, empathy and ethics. In the context of technology, these applications positively affect the perceptions of teachers, prospective teachers and students regarding their acceptance of artificial intelligence technologies, facilitate the learning of machine learning applications and provide quality education opportunities. Concerns about the use of technologies such as artificial intelligence stem from feelings of fear and insecurity as a result of not having enough information about the potential opportunities that these technologies will offer [19]. As explained within the framework of the Unified Theory of Acceptance and Use of Technology (UTAUT) model, performance expectation (AI tools making individuals' jobs easier) and effort expectation (not having to be long to use these tools) have a significant impact on individuals' technology acceptance and use status [50]. In the context of active participation, these applications have potential effects such as activating teacher-student and peer interactions and contributing to students' participation in interactive learning activities.

As a result of the systematic review, the findings regarding the potential effects of artificial intelligence applications in science and chemistry education as well as some risks/limitations are noteworthy. It is seen that the researchers are trying to minimize these risks/limitations with the suggestions they present. The studies reviewed generally report ethical issues such as gender and race bias, hallucinations, copyright infringement, plagiarism, and biased information production ($n = 3$). The existing literature reveals that research on artificial intelligence in education generally addresses the technical and pedagogical aspects of these technologies [1]. Researchers report that studies are needed to investigate the ethical effects of these technologies such as bias, injustice, and plagiarism. Researchers report that these effects in educational contexts can lead to racial and gender inequalities, leading to discrimination [3] and social hierarchy [18]. In addition, plagiarism detection in AI-generated content is difficult, which can lead to worrying consequences in terms of academic integrity [5]. Researchers stated issues related to accuracy and reliability ($n = 2$) as copying the information produced without verifying its accuracy and excessive trust in AI tools. Mao, Chen, and Liu [38] emphasize that students may over-rely on such tools to complete all their tasks, such as writing assignments, and this may have negative effects on writing and critical thinking skills. Researchers stated issues related to technical infrastructure and language support ($n = 2$) as difficulties in perceiving natural language expressions and language incompatibilities in software and interfaces. Researchers stated issues related to data dependency and model limitations ($n = 1$) as superficial analyses based on word statistics. Researchers stated that its impact on human decisions ($n = 1$) weakens decision-making processes. Similar to these findings, Sun and Hoelscher [46] report the limitations of artificial intelligence systems such as incorrect content production, inability to obtain up-to-date information, inability to provide insights based on personal experience and emotions, and limited word count production. Finally, regarding the risks/limitations of artificial intelligence applications in science and chemistry education, researchers stated that artificial intelligence negatively affects writing skills in terms of the effects on education and skill development ($n = 1$). Writing skills are a skill that can be developed with human interaction and personalized guidance [37]. Farrokhnia et al. [15] state that ChatGPT produces responses based on the datasets it was trained on and cannot provide individual feedback to students. Researchers see this as an obstacle to the development of certain writing skills in students. Some researchers emphasize that it may be important to investigate the potential benefits and challenges of ChatGPT's integration, especially in science education. The findings reveal that ethical issues such as bias, plagiarism, accuracy, and reliability affect the fair and reliable conduct of learning processes, while technical limitations weaken accessibility and user experience. Future researchers can take measures to support metacognitive skills such as writing, critical thinking, and decision-making in the use of artificial intelligence applications in science and chemistry education.

7 Conclusion and recommendations

As a result of the systematic review, studies focusing on the effects of artificial intelligence applications in science and chemistry education between the years 2014–2024 were examined. The findings obtained indicate that artificial intelligence applications are widely used especially in higher education institutions and this use has increased

between the years 2021–2024. While the primary target audience of teacher candidates indicates a strategic orientation towards increasing AI literacy of future practitioners, the limited number of studies focusing on high school and middle school levels indicates a significant gap in the integration of AI into early education levels. Increasing the number of studies, especially with high school and middle school student groups, can help to reveal the effects of these applications in educational contexts more clearly. When examined from a methodological perspective, the mixed research method is mostly used in the studies. This situation allows for a more comprehensive examination of the benefits and limitations of artificial intelligence applications in education. Including instructional design models in studies on artificial intelligence applications in education can more clearly reveal a systematic approach to the development, implementation and evaluation processes of artificial intelligence tools and applications. It is seen that various data collection methods and techniques such as surveys, scales, rubrics, achievement tests, videos and observation records are used to determine the effects of artificial intelligence tools and applications in science and chemistry education. This situation allows for more comprehensive analyses to be made by collecting multi-dimensional data in studies. Using different data collection techniques together in the data collection process can provide more reliable results.

This systematic review shows that AI applications provide significant learning outcomes in science and chemistry education and contribute to many areas from online learning processes to laboratory courses. While artificial intelligence tools and applications are widely used in science education, the use of these tools and applications in chemistry and STEM education is more limited. This situation reveals that the application diversity and research trend of artificial intelligence tools in science education is greater. Studies in the field of chemistry education focus more on technical applications. This imbalance reveals the need for pedagogical research addressing the integration of artificial intelligence into teaching processes in the context of chemistry education. Studies investigating the potential of artificial intelligence tools in chemistry and STEM education for the development of conceptual understanding, problem solving and laboratory skills can be increased. Projects/tasks can be developed to encourage the use of artificial intelligence technologies in chemistry and STEM education, and training can be organized to develop teachers' literacy skills in using these tools. ChatGPT is one of the artificial intelligence tools widely used in science and chemistry education and offers potential benefits such as personalized learning, feedback, increasing active participation, and reducing teachers' workload. In addition, it is seen that teachers benefit from artificial intelligence tools in evaluation processes. Studies aimed at developing students' artificial intelligence literacy will be valuable in terms of conscious use of these tools. It is seen that artificial intelligence tools positively affect learning processes in science and chemistry education. It is revealed that artificial intelligence tools and applications facilitate learning through personal assistant and smart feedback systems, increase self-confidence in writing skills, encourage active participation, and increase student–teacher–peer interaction; and that teachers develop professional and technological literacy skills, and support them in preparing lesson plans and developing different evaluation methods. In this context, training programs can be organized for teachers to encourage the effective and efficient use of artificial intelligence tools in science and chemistry education and awareness can be raised about the pedagogical use of these tools. There are some risks and limitations regarding the use of artificial intelligence tools in science and chemistry education. Among these limitations, gender and racial prejudices, plagiarism, hallucinations, accuracy and reliability problems, technical infrastructure and language support problems, and negative effects on writing skills are noteworthy. In addition, excessive reliance on the content produced by artificial intelligence tools can negatively affect students' independent thinking and decision-making skills. All these findings indicate that although the potential offered by artificial intelligence applications in education is great, the technology must be integrated correctly and ethically. In this context, it is important to determine the ethical issues regarding the use of artificial intelligence tools in education and present them to students and to increase the awareness of students and teachers about the conscious use of these tools.

Based on the findings of this systematic review, future research can delve deeper into the effects of AI applications in science and chemistry education. Experimental or mixed-method studies examining the effects of AI tools and applications on conceptual learning and science process skills at the high school and secondary school levels can be increased. Such studies can provide concrete findings on how AI facilitates learning, especially in courses with intensive abstract concepts such as science and chemistry. In addition, the effects of AI applications that support teacher-student-peer interaction on learning processes can be addressed in the context of teachers' professional development and technological pedagogical competencies. Such studies can make significant contributions to the effective pedagogical use of AI tools and applications.

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Data availability All articles analyzed during the current study are available in the academic databases Web of Science and Scopus. Query generated for search in Web of Science: <https://www.webofscience.com/wos/woscc/summary/5358a763-0978-427b-ba55-c8515f043713-0146f8f20d/relevance/1> Query generated for search in Scopus: TITLE-ABS-KEY ("science education" OR "chemistry education" AND "artificial intelligence" OR "generative artificial intelligence") AND PUBYEAR > 2013 AND PUBYEAR < 2025 AND (LIMIT-TO (SUBJAREA, "SOC")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English")).

Declarations

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